

2.2 Group 2

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.2 Group 2
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Solids **Q**, **R**, **S** and **T** are compounds of two different Group II metals. A few of the uses of these are detailed below.

Compound **Q** is used as a refractory lining material in kilns.

Compound **R** is used as a building material. It can also be heated in a kiln to form compound **S**. When **S** is hydrated; it forms compound **T**, which is used agriculturally to treat soils.

Which statements about these compounds are correct?

- 1 More acid is neutralised by 1.0 g of **R** than by 1.0 g of **Q**
- 2 The M_r of **R** is greater than the M_r of **S** by 44.0
- 3 The metallic element in **S** reacts with cold water more rapidly than the metallic element in **Q**

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 2

An anhydrous nitrate of a Group II metal was thermally decomposed. 6.00 g was used, and 3.06 g of gas was produced.

Use of the periodic table is relevant to this question.

What is the nitrate compound?

- A. Beryllium nitrate
- B. Strontium nitrate
- C. Magnesium nitrate
- D. Calcium nitrate

[1 mark]

Question 3

A mixture of 2 Group II elements, **Y**, were thermally decomposed.

This reaction produced a white solid and only two gaseous products. One of the gaseous products relights a glowing splint.

What could be the components of mixture **Y**?

- A. MgCl_2 and CaCO_3
- B. MgCO_3 and $\text{Ca(NO}_3)_2$
- C. MgO and CaSO_4
- D. $\text{Mg(NO}_3)_2$ and $\text{Ca(NO}_3)_2$

[1 mark]

Question 4

Use of the periodic table is relevant to this question.

Thermal decomposition of anhydrous magnesium nitrate, $\text{Mg(NO}_3)_2$, gives a white solid and a mixture of two gases **R** and **S**.

S is oxygen.

What is the ratio $\frac{\text{mass of } \mathbf{R} \text{ released}}{\text{mass of } \mathbf{S} \text{ released}}$?

- A. $\frac{1}{0.267}$
- B. $\frac{1}{0.174}$
- C. $\frac{1}{0.348}$
- D. $\frac{1}{3.43}$

[1 mark]

Question 5

When equal volumes of saturated solutions of barium hydroxide and calcium hydroxide are mixed, a white precipitate, **Y**, forms. The mixture is filtered, and carbon dioxide is bubbled through the filtrate, producing a second white precipitate, **Z**.

What are **Y** and **Z**?

	Y	Z
A	$\text{Ba}(\text{OH})_2$	$\text{Ca}(\text{OH})_2$
B	$\text{Ba}(\text{OH})_2$	CaCO_3
C	$\text{Ca}(\text{OH})_2$	BaCO_3
D	$\text{Ca}(\text{OH})_2$	$\text{Ba}(\text{OH})_2$

[1 mark]

Question 6

A colourless solution was formed when metal **Y** was reacted with water. This solution then gave a white precipitate when mixed with aqueous sulfuric acid.

What is metal **Y**?

- A. Potassium
- B. Magnesium
- C. Barium
- D. Sodium

[1 mark]

Question 7

A student performs several experiments on element Y, an alkaline earth metal to determine its identity and the following observations are made:

- 1 When added to water, Y-hydroxide dissolves in water.
- 2 When added to water, Y-sulfate produces a white precipitate.
- 3 In order to thermally decompose Y-nitrate, more heat is required than to thermally decompose strontium nitrate.
- 4 When burned in oxygen Y will readily become YO.

What is the identity of element Y?

- A. Beryllium
- B. Barium
- C. Calcium
- D. Magnesium

[1 mark]

Question 8

Group II metal carbonates can be used as fire retardants as they thermally decompose producing CO₂. The CO₂ removes the oxygen from the fire. Three minerals each having a mixture of two Group II metal carbonates could be used as fire retardants.

Barytocite BaCa(CO₃)₂

Dolomite CaMg(CO₃)₂

Huntite Mg₃Ca(CO₃)₄

What is the order of effectiveness as a fire retardant, from best to worst?

	Best	— — — — — →	Worst
A	Dolomite	Barytocite	Huntite
B	Dolomite	Huntite	Barytocite
C	Huntite	Dolomite	Barytocite
D	Huntite	Barytocite	Dolomite

[1 mark]

Question 9

Thermal decomposition of the nitrates of beryllium, calcium, magnesium, and strontium produce the same products. When 4.00 g of one of these anhydrous nitrates is decomposed, 2.64 g of gas is produced.

Use of the periodic table is relevant to this question.

What is the nitrate?

- A. Beryllium nitrate
- B. Magnesium nitrate
- C. Calcium nitrate
- D. Strontium nitrate

[1 mark]

Question 10

In which row of the table are all statements comparing magnesium and barium correct?

	First ionisation energy of magnesium	First ionisation energy of barium	Reaction of magnesium with oxygen	Reaction of barium with oxygen
A	lower	higher	faster	slower
B	lower	higher	slower	faster
C	higher	lower	faster	slower
D	higher	lower	slower	faster

[1 mark]

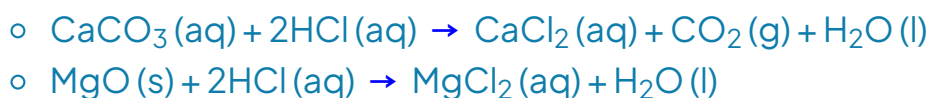
Model Answers: Hard

1

The correct answer is **C** because:

- The question states that:
 - Compound **Q** is used as kiln lining.
 - The Group II metal compound with the highest melting point is **magnesium oxide**.
 - Compound **R** is used in building materials, so this is **calcium carbonate** (limestone).
 - When compound R is heated it forms **S**, this will be thermal decomposition and produces **calcium oxide**.
 - When calcium oxide is hydrated, it produces **calcium hydroxide (T)** that is used to increase the pH of the soil in agriculture.
- The correct statements will be 2 and 3 because:
 - The M_r of calcium carbonate is (R) 100, the M_r of calcium oxide is 66 as the M_r of carbon dioxide is 44.
 - Calcium is more reactive in cold water than magnesium.

Statement 1 is incorrect as they would have the same neutralisation effect on acids as each other. One mole of calcium carbonate reacts with hydrochloric acid as does one mole of magnesium oxide.



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The correct answer is **B** because:

- Decomposition of anhydrous Group II nitrates-
 - $2\text{X}(\text{NO}_3)_2 \rightarrow 2\text{XO} + 4\text{NO}_2 + \text{O}_2$, where X is the Group II metal.
- There are two different gases produced. Therefore, it is not possible to find the

number of moles of the gases produced with the information given.

- **Amount of solid metal oxide** = $6.00 - 3.06 = 2.94 \text{ g}$
- From the equation, 2 moles of the metal nitrate will produce 2 moles of the metal oxide. Hence it is a 1 : 1 mole ratio.
- In order to solve this question, the molecular mass of each metal nitrate and metal oxide must be calculated. The answer would be the compound which has an equal number of moles of metal nitrate and metal oxide for the masses given.
- **Number of moles strontium nitrate**
 - $\text{moles} = \frac{\text{mass}}{M_r} = \frac{6}{211.6} = 0.028 \text{ mole}$
- **Number of moles strontium oxide**
 - $\text{moles} = \frac{\text{mass}}{M_r} = \frac{2.94}{103.6} = 0.028 \text{ mole}$
- Hence **strontium nitrate** is the compound.

3

The correct answer is **D** because:

- The question states that **Y** is a mixture of two compounds.
- One of the gases produced from thermal decomposition relights a glowing splint, so this must be oxygen.
- The compounds must have been nitrates.
- Decomposition of anhydrous Group II nitrates–
 - $2X(\text{NO}_3)_2 \rightarrow 2X\text{O} + 4\text{NO}_2 + \text{O}_2$, where X is the Group II metal.
- The combination of compounds that would produce oxygen is option **D**.

A is incorrect as magnesium chloride and calcium carbonate would not produce oxygen as a product when thermally decomposed.

B is incorrect as magnesium carbonate and calcium nitrate would produce three gaseous products; carbon dioxide gas, oxygen gas and nitrogen dioxide.

C is incorrect as magnesium oxide is often used as fireproofing so will not readily decompose, and calcium sulfate would not produce oxygen as a product unless temperatures were extremely high at 1350 – 1400 °C.

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The correct answer is **B** because:

- Decomposition of anhydrous Group II nitrates-
 - $2X(NO_3)_2 \rightarrow 2XO + 4NO_2 + O_2$, where X is the Group II metal.
- The two gases produced are nitrogen dioxide and oxygen.
- 1 mole of oxygen gas is produced when 4 moles of nitrogen dioxide gas.
- 1 mole of oxygen has a mass of 32 g.
- 4 moles of nitrogen dioxide gas has a mass of 184 g.

- The ratio of $\frac{\text{mass of R released}}{\text{mass of S released}} = \frac{184}{32} = \frac{\frac{184}{184}}{\frac{32}{184}} = \frac{1}{0.174}$

5

The correct answer is **C** because:

- The solubility of the Group II hydroxides increases going down Group II.
- Barium hydroxide is more soluble than calcium hydroxide; the barium hydroxide will dissociate more in solution giving a higher concentration of hydroxide ions.
- When the two solutions are added together, the calcium hydroxide will precipitate,

The correct answer is **C** because:

- The solubility of the Group II hydroxides increases going down Group II.
- Barium hydroxide is more soluble than calcium hydroxide; the barium hydroxide will dissociate more in solution giving a higher concentration of hydroxide ions.
- When the two solutions are added together, the calcium hydroxide will precipitate, this is removed by filtering.
- When the barium hydroxide solution (the filtrate), has carbon dioxide bubbled through it will react to produce barium carbonate.
 - $\text{Ba(OH)}_2 + \text{CO}_2 \rightarrow \text{BaCO}_3 + \text{H}_2\text{O}$

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The correct answer is **C** because:

- The equations for these reactions are
 - $\text{Ba(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Ba(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$
 - $\text{Ba(OH)}_2\text{(aq)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)} + 2\text{H}_2\text{O(l)}$
- The solution formed when barium was reacted with water was barium hydroxide, as barium hydroxide is more soluble the solution would be colourless.
- When mixed with sulfuric acid, a neutralisation reaction takes place producing an insoluble precipitate of BaSO_4 .

A is incorrect as potassium would make a colourless solution of potassium hydroxide but when mixed with sulfuric acid the salt produced would be soluble.

B is incorrect as magnesium would give a slightly cloudy solution of magnesium hydroxide and when mixed with sulfuric acid the salt produced would be soluble.

D is incorrect as sodium would make a colourless solution of sodium hydroxide but when mixed with sulfuric acid the salt produced would be soluble.

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The correct answer is **B** because:

- Group II metal hydroxides are more soluble in water as you go down the group.
 - Barium hydroxide will dissolve readily in water.
- Group II metal sulfates become less soluble as you go down the group.
 - Barium sulfate is more likely to produce a precipitate in water.
- The heat required to thermal decompose a Group II metal nitrate increases as you go down the group.
 - Barium nitrate will need a higher temperature than strontium nitrate.
- The reactivity in oxygen increases as you go down the group.
 - Barium will react readily with oxygen and is stored under oil to prevent this reaction.

A, C & D are incorrect as they are all above barium in Group II so will have less solubility as a nitrate, more solubility as sulfate and less energy needed to thermally decompose than strontium.

8

The correct answer is **C** because:

- The effectiveness of the compounds as fire retardants depends on the volume of CO_2 produced.
- The volume of CO_2 is determined by how easy the Group II compound is to decompose.
- As you go down Group II the thermal stability increases so, more energy would be required to decompose the carbonate.
- The order of the metals shown in the question from top to bottom is magnesium, calcium and then barium.
- The best fire-retardant compounds will be made of metal carbonates higher up the group so huntite would be the best as there is a higher proportion of magnesium meaning this will decompose more readily under the same conditions.
- Then dolomite as there is an equal proportion of magnesium and calcium in this compound.

- The last one, therefore, would be the barytocite as it contains barium, which is lower down the group.

9

The correct answer is **C** because:

- Decomposition of anhydrous Group II nitrates–
 - $2X(NO_3)_2 \rightarrow 2XO + 4NO_2 + O_2$, where X is the Group II metal.
- There are two different gases produced. Therefore, it is not possible to find the number of moles of the gases produced with the information given.
- **Amount of solid metal oxide** = $4.00 - 2.64 = 1.36$ g.
- From the equation, 2 moles of the metal nitrate will produce 2 moles of the metal oxide. Hence it is a 1 : 1 mole ratio.
- In order to solve this question, the molecular mass of each metal nitrate and metal oxide must be calculated. The answer would be the compound which has an equal number of moles of metal nitrate and metal oxide for the masses given.
- **Number of moles calcium nitrate**
 - $moles = \frac{mass}{M_r} = \frac{4}{164} = 0.024 \text{ mole}$
- **Number of moles calcium oxide**
 - $moles = \frac{mass}{M_r} = \frac{1.36}{56} = 0.024 \text{ mole}$
- Hence **calcium nitrate** is the compound

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- **Number of moles calcium nitrate**

- $\text{moles} = \frac{\text{mass}}{M_r} = \frac{4}{164} = 0.024 \text{ mole}$

- **Number of moles calcium oxide**

- $\text{moles} = \frac{\text{mass}}{M_r} = \frac{1.36}{56} = 0.024 \text{ mole}$

- Hence **calcium nitrate** is the compound

10

The correct answer is **D** because:

- Group II elements get more reactive as you go down the group.
- **As you go down the group the first ionisation energy of Group II decreases.**
- There are more filled shells between the nucleus and the outer electrons; this shields the outer electrons from the attraction of the nucleus.
- The radius of the atom increases, so the distance between the nucleus and the outer electrons increases.
 - Therefore, the force of attraction between the nucleus and outer electrons is reduced.
 - So, less energy is needed to remove an outer electron.
- **As you go down the group, the reaction with oxygen is more vigorous.**
 - Magnesium burns with a bright white light.
 - Barium is so reactive it is stored under oil to keep it out of contact with air.

